AKASH JYOTI HANDIQUE

Energy Systems Researcher

Christchurch, New Zealand | +64 27 303 5413 | akashjyoti.handique@pg.canterbury.ac.nz | Website: akashhandique.github.io

PROFESSIONAL SUMMARY

Energy systems researcher with experience building and applying optimisation-based capacity-expansion models to renewable electricity transition planning across multiple energy carriers, including power, heat, transport, and industry. Experienced in multi-country collaboration across New Zealand, Germany, Chile, and Finland, and in translating technical model outputs for government ministries and regulators.

CORE COMPETENCIES

- Techno-economic and cost analysis of renewable power generation and storage technologies
- Renewable energy database compilation, verification, and ongoing maintenance
- Capacity-expansion and generation-mix optimisation modelling (REMix, PyPSA-Earth, HOMER)
- Data visualisation and technical reporting for multidisciplinary and international audiences
- Renewable energy technology and supply chain literature synthesis

PROFESSIONAL EXPERIENCE

Research Assistant, SERG — *University of Canterbury, New Zealand · Jun 2023 - Present*

- Compiled and now maintain an open-access energy demand and renewable supply dataset for the Pacific Islands, later published as a peer-reviewed data descriptor (EDRS-PIC, IEEE Data Descriptions, 2026)
- Completed a research visit to the German Aerospace Centre (DLR) in Stuttgart, Germany, to develop the REMix-PIC energy transition model in collaboration with DLR's Energy Systems Analysis team
- Contributed technical analysis to four government and regulatory consultations
- Supervised undergraduate and postgraduate students on thesis projects in renewable energy systems

Student Assistant, Institute for Modelling Hydraulic and Environmental Systems — *University of Stuttgart, Germany · May 2021 - Oct 2022*

- Processed geospatial data in QGIS to support river and floodplain modelling for an international sediment-transport research collaboration with MIT, conducted in Bavaria, Germany
- Ran hydro-morphodynamic numerical models and characterised sediment properties through laboratory testing, including water displacement and sieve analysis

POLICY AND REGULATORY CONTRIBUTIONS

- Electricity Authority NZ, “Improving Visibility of Generation Investment: Clause 2.16 Information Notices,” feedback submission, University of Canterbury, 2024. Link
- Ministry for the Environment NZ, “New Zealand's Second Emissions Reduction Plan,” feedback report, University of Canterbury, 2024.
- Ministry of Business, Innovation and Employment NZ, “Interim Hydrogen Roadmap,” feedback report, University of Canterbury, 2023. Link
- Ministry of Business, Innovation and Employment NZ, “A Proposed Reliability Obligation to Manage Dry-Year Risk,” consultation, SERG, 2026. Link

EDUCATION

Ph.D. Candidate, Civil and Environmental Engineering — *University of Canterbury, New Zealand · 2023 - 2026 (expected)*
Thesis: Energy Transitions for the Pacific Islands

M.Sc., Water Resources Engineering and Management — *University of Stuttgart, Germany · 2020 - 2022*

B.Tech., Civil Engineering — *Tezpur University, India · 2015 - 2019*

PAPERS

Handique, A.J., Peer, R., Haas, J. et al. (2024). *Distributed hydrogen systems: A literature review (synthesis of 159 studies).* International Journal of Hydrogen Energy, 85, 427-439. DOI: 10.1016/j.ijhydene.2024.08.206

Handique, A.J., Peer, R.A.M., Haas, J. (2024). *Understanding the Challenges for Modelling Islands' Energy Systems and How to Solve Them.* Current Sustainable/Renewable Energy Reports. DOI: 10.1007/s40518-024-00243-8

Handique, A.J. et al. (2025). *Characterising Energy Models for Island Grids: HOMER vs. PyPSA-Earth in Fiji's 2030 Scenario.* IEEE PES Innovative Smart Grid Technologies (ISGT) Europe. DOI: 10.1109/ISGTEurope64741.2025.11305505

Handique, A.J. et al. (2026). *Descriptor: Energy Demand and Renewable Supply Dataset for Pacific Island Countries (EDRS-PIC).* IEEE Data Descriptions. DOI: 10.1109/IEEEDATA.2026.3685952

Handique, A.J., Peer, R., Haas, J., Sofe, R. (2026). *Beyond Diesel: 100% Renewable Electricity Pathways for Pacific Island Countries under Land and Resource Constraints.* SSRN preprint. SSRN Abstract 7305262

Handique, A.J. et al. (2026). *From Imported Fuels to Energy Independence: A Net-Zero Transition Pathway for Pacific Island Countries by 2050.* SSRN Abstract 7540170

Sepúlveda-Figueroa, V. et al., including **Handique, A.J.** (2026). *New Critical Design Conditions: Event-Aware Planning of Isolated Renewable Power Systems.* (Working paper)

PRESENTATIONS

Handique, A.J., Breyer, C., Lopez, G., Haas, J., Peer, R. *A Comprehensive Energy Dataset for the Pacific Region.* Department of Energy Systems Analysis, German Aerospace Centre (DLR), Stuttgart, Germany, Dec 2025.

Handique, A.J., Vatankhah Ghadim, H., Zegeer, M.L., Peer, R., Haas, J. *Characterising Energy Models for Island Grids: HOMER vs. PyPSA-Earth in Fiji's 2030 Scenario.* IEEE PES Innovative Smart Grid Technologies (ISGT) Europe, Valletta, Malta, Oct 2025.

Handique, A.J., Haas, J., Peer, R. *Power the Pacific: Energy Transitions in Island Nations.* 20th International Conference on Sustainable Development of Energy, Water and Environment Systems (SDEWES), Dubrovnik, Croatia, 2025.

Handique, A.J., Haas, J., Peer, R. *Electricity Transitions for the Overlooked: 14 Pacific Island Countries.* 2nd International Symposium on Energy Systems Analysis (ISESA), Stuttgart, Germany, 2025.

Handique, A.J., Haas, J., Peer, R. *100% Renewable Electricity Transition Pathways for the Pacific Islands.* 3rd New Zealand Hydrogen Symposium (NZHS 2025), Christchurch, New Zealand, Feb 2025.

Handique, A.J., Haas, J., Peer, R., Osorio-Aravena, J.C., Reyes-Chamorro, L. *What is the Role of Distributed Hydrogen Systems in Island Energy Systems?* 2nd New Zealand Hydrogen Symposium (NZHS 2024), Wellington, New Zealand, Feb 2024.

OPEN-SOURCE RESEARCH TOOLS

REMix-PIC, Pacific Island Countries Energy Transition Model (2026, latest). GitHub

GUI for Hydrograph Extrapolation. GitHub

Spatially Explicit, Depth-Dependent, Data-Driven Roughness Calibration for Numerical Modelling. GitHub

TECHNICAL SKILLS

Modelling and Analysis: REMix, PyPSA-Earth, HOMER, GAMS, Python, R, MATLAB, EnergyPLAN

Other: QGIS, TELEMAC, MODFLOW, EPANET, AutoCAD, Microsoft Office

Languages: English (C1) - German (B1) - Hindi (C2) - Assamese (C2, native)

AWARDS

- Zweckverband Landeswasserversorgung Master's Scholarship, Germany, 2021
- Ishan Uday Award, University Grants Commission, India, 2015
- Anundoram Borooah Award, Government of India, 2013